ANDRY PAEZ

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EXPERIENCE

Research Assistant, Dept. of Physics and Astronomy **Feb 2026 – Present**

San Jose State University (advisor: Dr. Hilary Hurst), 30 hrs/week

- Develop spinor Bose-Einstein condensate simulations (Gross-Pitaevskii equation); all work merged into the research group's codebase through advisor-reviewed pull requests.
- Vectorized the core kinetic-energy operator for up to 1,880x speedup (161 s to 0.09 s) with bit-exact numerical parity verified by a benchmark harness.
- Built the data pipeline: CLI with HDF5 output, hash-addressed run catalog with rebuildable SQLite index, and a Slurm write-then-index HPC workflow that avoids concurrent-writer contention.
- Designed controlled storage benchmarks and authored the store-vs-recompute recommendation adopted as the project default.

SKILLS

- **AI Systems:** local LLM inference (llama.cpp, Ollama), agent architectures and tool-calling loops, RAG, MCP, edge deployment (Jetson Orin Nano), faster-whisper; multi-agent coding orchestration with issue-driven workflows and verification-first review gates
- **LLM Internals:** quantization formats (K-quants, importance-matrix I-quants) and quantization-aware model selection, LoRA fine-tuning, KV-cache memory/compression tradeoffs, MoE expert pruning
- **Languages:** Python (primary), TypeScript/JavaScript, Rust, C/C++, SQL, Bash
- **Systems:** FastAPI, WebSockets/SSE, Tauri, pytest, GitHub Actions CI, Docker, Playwright, Linux (systemd, GStreamer, ALSA/PipeWire), Tailscale, Git

PROJECTS

Reachy Mini Robot Control Console **Jun 2026 – Present**

- Built a FastAPI web control console for a Reachy Mini robot on a headless Jetson Orin Nano, designed as a pure HTTP/WebSocket proxy so the robot daemon stays the single writer to the hardware; functionally verified against the live robot.
- Built an in-place app-library voice pipeline from local faster-whisper STT through the Hermes agent to neural TTS, including live camera frames for visual questions; persistent sessions and access-controlled native-session reuse enabled cancellation and queued follow-ups, with live Jetson validation confirming camera sharing continued while Hermes ran.
- Root-caused failures across the stack: compiled GStreamer webrtcsink from source for arm64 with NVMM buffer transport, traced speaker silence to ALSA card remapping after an OS migration, and fixed a SIGINT-forwarding bug with a process-tree signaller.
- Reduced idle UI overhead to zero DOM writes per tick (change-guarded rendering, SSE rate reduction), fixing a measured phone battery/heat problem; verified by driving the served app with Playwright.

LocalLM, Agentic Desktop App for Local Models **Apr 2026**

- Built a cross-platform desktop app for local Ollama models with an agentic tool-calling layer (web search, file I/O, code execution) in Rust (Tauri 2) with a React/TypeScript frontend and streaming chat.
- Implemented a model recommendation registry with RAM/disk metadata and machine-spec detection for quantization-aware model selection.

omnisense, Offline Voice and Vision Assistant for Jetson **Apr 2026**

- Built a modular multimodal assistant package: camera abstraction (GStreamer/OpenCV/mock backends), faster-whisper speech-to-text, voice activity detection, YOLO overlay, FastAPI REST/WebSocket interface; testable without hardware via mock backends.

Sage, Local-First Knowledge Agent **Feb 2025 – May 2026**

- Designed and built a desktop knowledge agent sustained over 15 months: document ingestion into a local SQLite store with embeddings, grounded chat with citations, quiz/report/audio generation; Tauri v2, Next.js, FastAPI, multi-provider LLM abstraction (OpenAI, Anthropic, Ollama), CI.

OPEN SOURCE

- **NousResearch/hermes-agent:** merged fixes for a systemd gateway restart race and Electron main-window lifecycle recovery in a large open-source agent framework.

EDUCATION

B.S., Physics **May 2026**

San Jose State University. GPA 3.80. Dean's Scholar (Fall 2024, Fall 2025)

Relevant Coursework: Computational Methods in Physics, Data Structures and Algorithms, UNIX/Linux Systems, Networking, Quantum Information Science